Divide and Inform:
Carving Up a Common Value Among Rival Bidders

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Information Market for Lending

- Competing lenders buy external information used for screening borrowers:
  - underlying data characteristics,
  - distilled prediction scores
- Information providers selling data:
  - Experian, Equifax, TransUnion – credit histories, credit scores
  - Enigma, Nova Credit – business revenues, consumer cash flows
- Lenders develop internal models to predict credit risk:
  - Underlying data is common across lenders
  - Models developed independently weigh common characteristics differently
  - Building an independent prediction model is expensive and its quality is limited by the lender's scope and size

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Key observation:

- A common provider can shape how much competing lenders agree about the same borrower

This Paper

- How does an information seller design information for competing lenders?
  - Whom to inform, what each lender learns, and how much to charge
- Does customization sharpen or soften lending competition?
  - Implications for seller revenue, lending profits, and borrower surplus
- How does the intensity of competition affect the information seller's ability to customize and charge for information?
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  - Lending competition vs. information provider competition
- Our approach
  - Lenders compete for a borrower of uncertain quality in a first-price auction
  - An information seller chooses the precision and correlation of their private signals

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3. **Strategic Segmentation:** a score can reveal more than credit risk
   - can inform about the intensity of competition
   - carve out a core group of high-quality borrowers for each lender
   - faced with weaker competition for core borrowers, lenders can charge higher rates

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Competition in lending and information markets has different effects:

- More lenders: lowers information seller's profits and increases borrower surplus
- More information sellers: lowers information fees but leaves borrower welfare unchanged

Related Literature

- Selling information to competitors
  Admati and Pfleiderer (1986, 1990), Bimpikis et al. (2019), Bonatti et al. (2024)
  Our contribution: negative correlation softens competition for a common-value loan

- Lending competition under asymmetric information
  Broecker (1990), Hauswald and Marquez (2003, 2006), He et al. (2023), Koung et al. (2026)
  Our contribution: the division of information across lenders is a choice of the information seller

- Information design in first-price auctions
  Bergemann and Morris (2016), Bergemann et al. (2017), Elliott et al. (2026)
  Our contribution: competition for risky loans vs. cash auctions

- Hub-and-spoke collusion sustained by a mediator
  Clark et al. (2024), Ortner et al. (2024), Sugaya and Wolitzky (2018, 2026)
  Our contribution: incentives via information vs. dynamic punishments

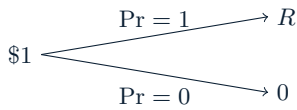
Model

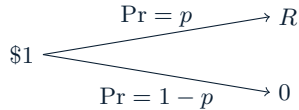
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  - requires $1 to fund a project

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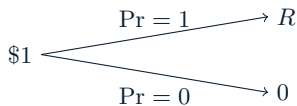
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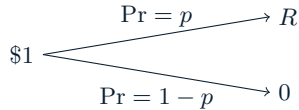


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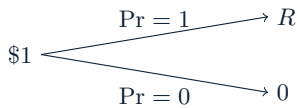
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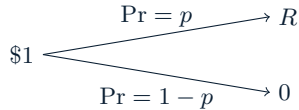
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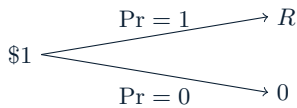
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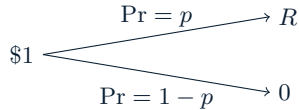
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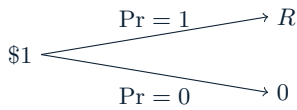
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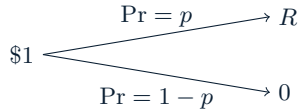
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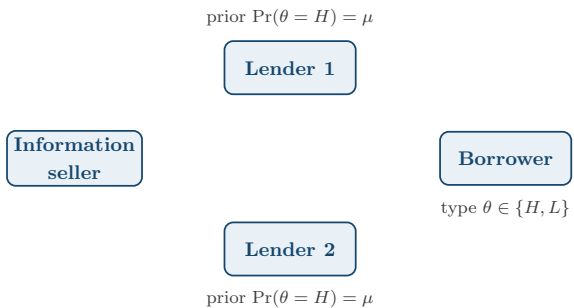
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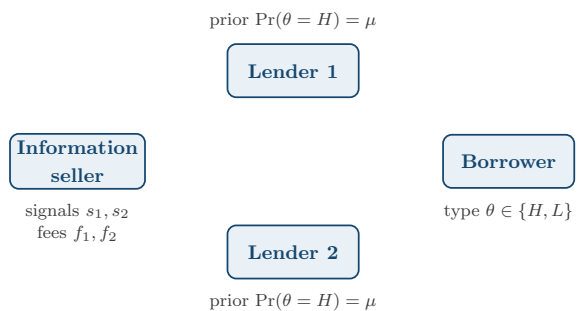
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  - charges fees f_1, f_2

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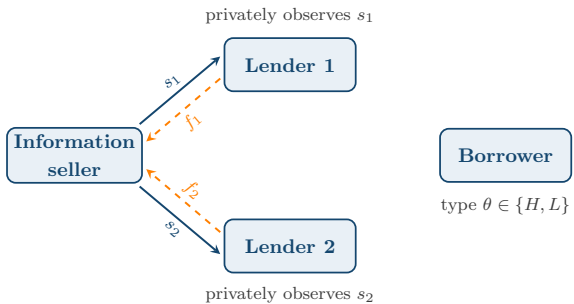
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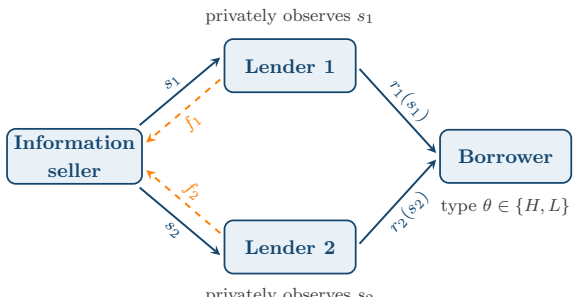
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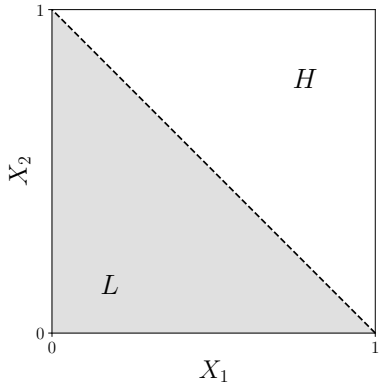
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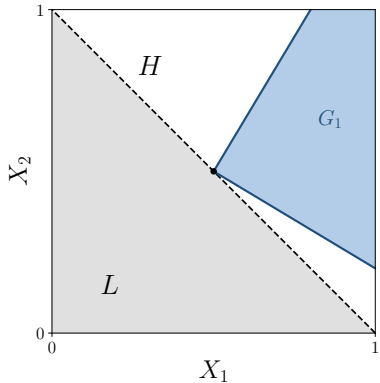
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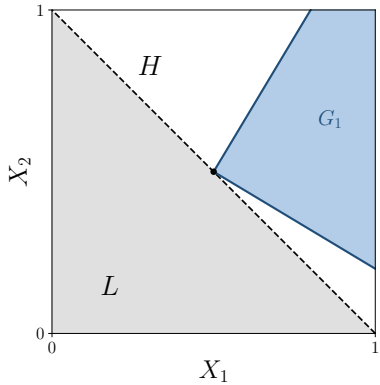
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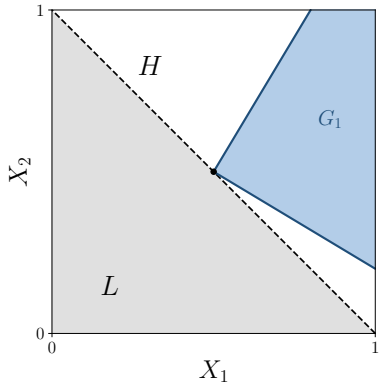


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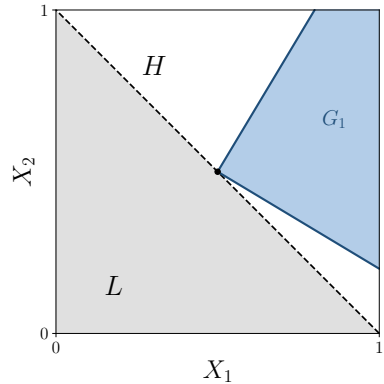


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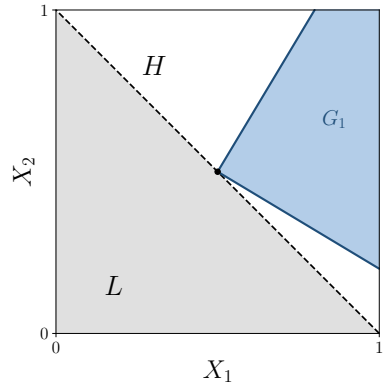


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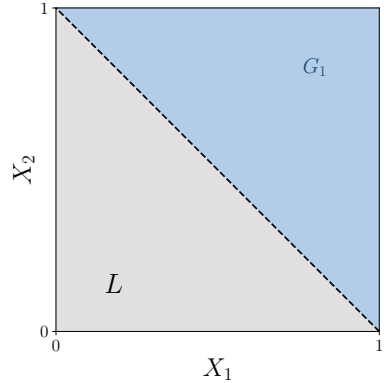


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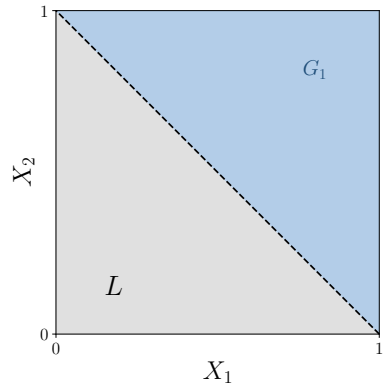
Proposition

When informing a single lender, the seller optimally supplies a perfectly revealing signal, $\alpha^* = 1$, and earns

$$\mu\big(\min\{R, \underline{R}(\mu)\} - 1\big).$$

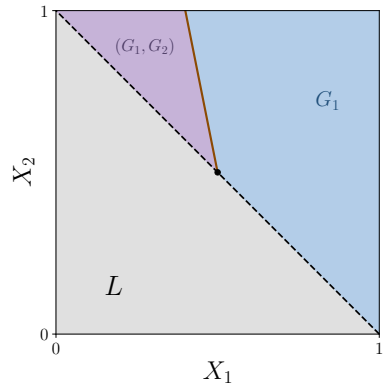
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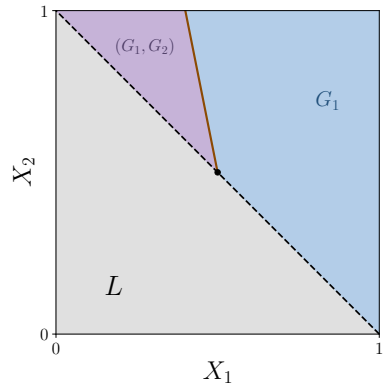
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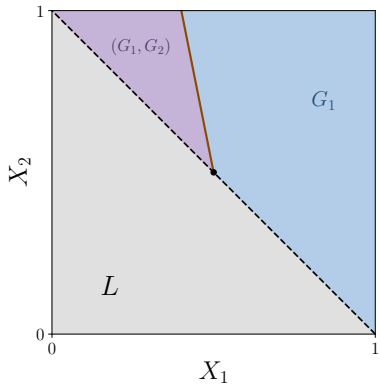
- Grade B discourages lending

$$\mu_B(\alpha_2) = \frac{(1-\alpha_2)\mu}{(1-\alpha_2)\mu + (1-\mu)} < \mu \;\Rightarrow\; \underline{R}(\mu_B) > \underline{R}(\mu)$$

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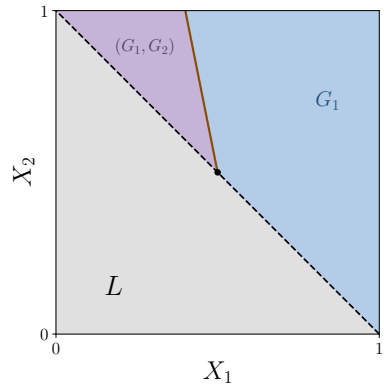


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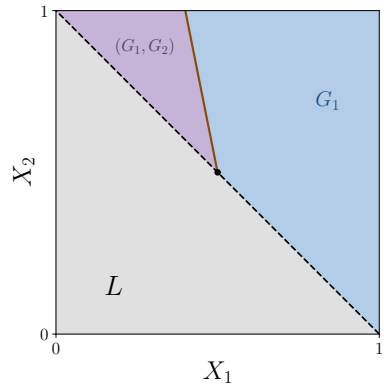


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- **Deterrence** of Lender 2 with B signal increases joint profits
  - Lender 1 faces a competitor only when $s_2 = G \Rightarrow$ competes less aggressively
  - Lender 2 with G signal faces less aggressive lender 1 $\Rightarrow$ competes less aggressively

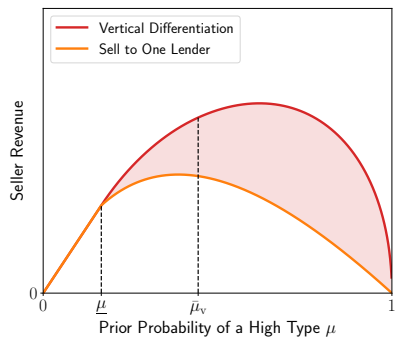
Optimal Vertically Differentiated Information

- Information is vertically differentiated if every high-type borrower recognized by lender 2 is also recognized by lender 1
  - Signals with $\alpha_1 \geq \alpha_2$ and

$$\mathrm{P}(s_1 = \mathsf{G} \mid s_2 = \mathsf{G}) = 1$$

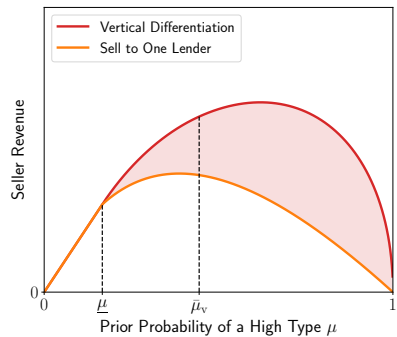
Optimal Vertically Differentiated Information

- Information is vertically differentiated if every high-type borrower recognized by lender 2 is also recognized by lender 1
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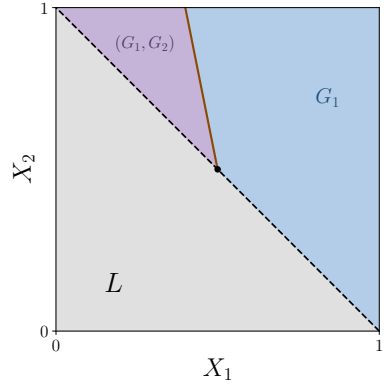
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- Lender 2's precision α_2:
  - $\mu \leq \underline{\mu}$: $\alpha_2^* = 0$ and lender one is a monopolist
  - $\mu \in (\underline{\mu}, \bar{\mu}_{\mathrm{v}})$: lowest precision α_2^* that deters entry after B
  - $\mu > \bar{\mu}_{\mathrm{v}}$: deterrence is too costly, and $\alpha_2 > 0$ only discourages lending after B



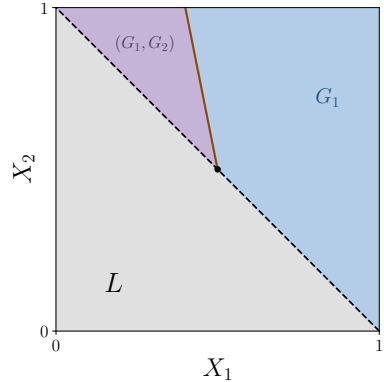
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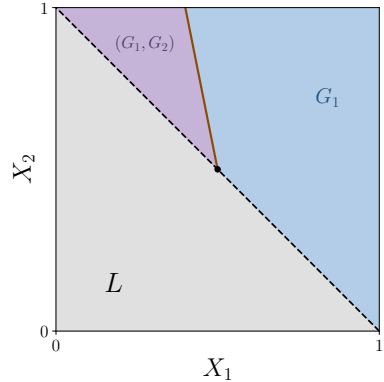
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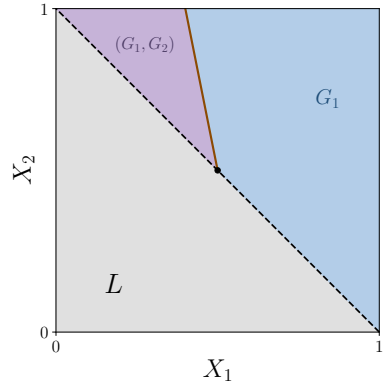
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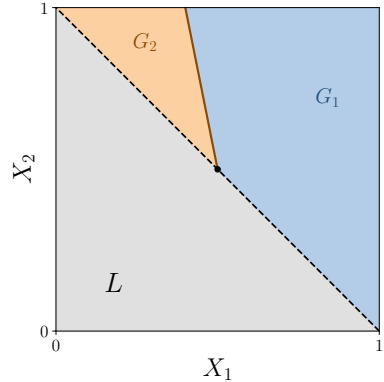
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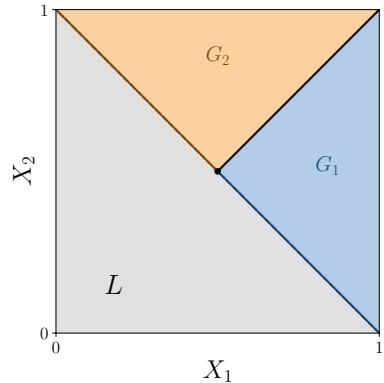
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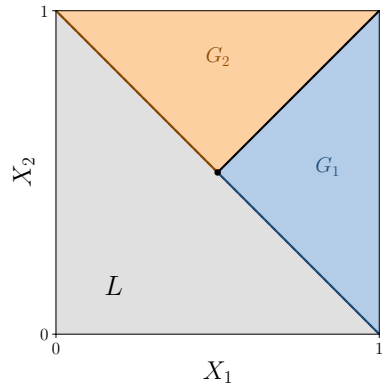
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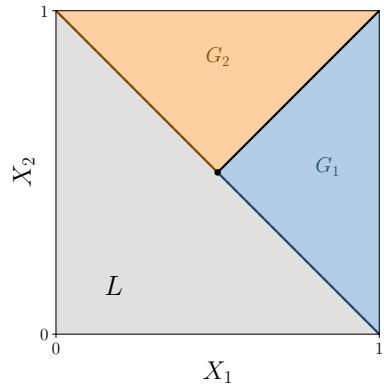
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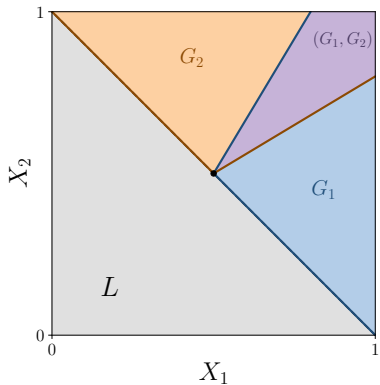
Deterrence vs. Segmentation

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- Higher precision can restore deterrence but it increases competition between lenders who observe G grades
- With $\alpha > 1/2$ a share $q > 0$ of high-type borrowers is contested

$$q = \mathrm{P}(s_{-i} = G \mid s_i = G) \geq q_{\min}(\alpha) = \frac{2\alpha - 1}{\alpha},$$

lenders bid rates and per-loan profit is

$$\underline{R}_G(q) - 1 = (R - 1)(1 - q)$$



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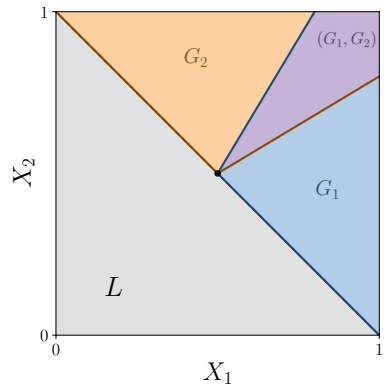
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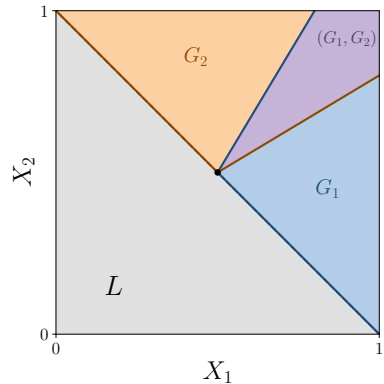
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lenders bid rates and per-loan profit is

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- B grade creates deterrence if $\underline{R}(\mu_B) \geq \underline{R}_G(q)$
- Otherwise lowers G's per-loan profits to

$$\underline{R}(\mu_B) - 1$$



Optimal Horizontally Differentiated Information

Proposition

When $\mu \leq \bar{\mu}$

- $q^* = 0$ perfect segmentation
- $\alpha^* = 1/2$ sufficient for deterrence

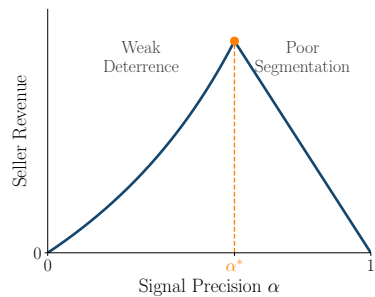
When $\mu > \bar{\mu}$

- $q^* = q_{\min}(\alpha^*)$ maximal segmentation
- $\alpha^* > 1/2$ ensures deterrence via

$$\underline{R}_G(q_{\min}(\alpha^*)) = \underline{R}(\mu_B(\alpha^*)).$$

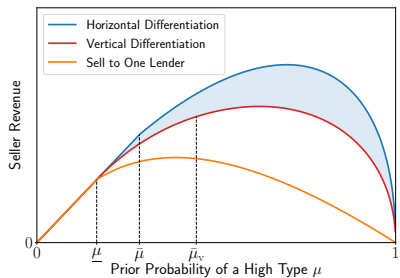
The information seller's profits can be written as

$$\mu(R-1)\big[1 - \mathrm{P}(G_1, G_2 \mid H)\big]$$



Value of Horizontal Differentiation

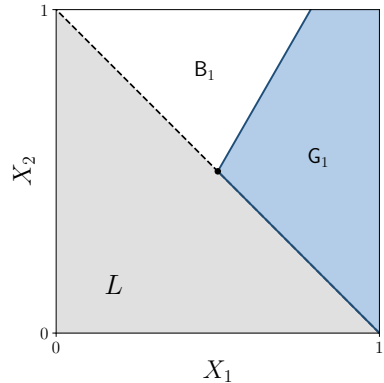
- When $\mu \leq \underline{\mu}$ even an uninformed lender does not lend
  - Monopoly profits regardless of what information is sold
- When $\mu \in (\underline{\mu}, \bar{\mu})$ precision $\alpha^* = 1/2$ deters entry without any competition
  - Perfect segmentation achieves monopoly profits
  - Vertically differentiated information creates deterrence at a cost
- When $\mu \in (\bar{\mu}, \bar{\mu}_{\mathrm{v}})$ precision $\alpha^* > 1/2$ deters entry with minimal competition
  - Segmentation softens competition
- When $\mu > \bar{\mu}_{\mathrm{v}}$ precision $\alpha^* > 1/2$ deters entry
  - Vertically differentiated signal only discourages lending after B



Optimal Information Design

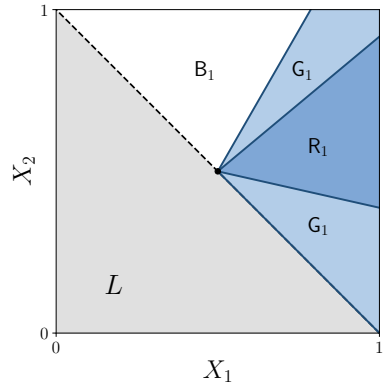
Differentiated Good News Signals

- Grade G reveals the borrower type but leaves the lender uncertain about competition



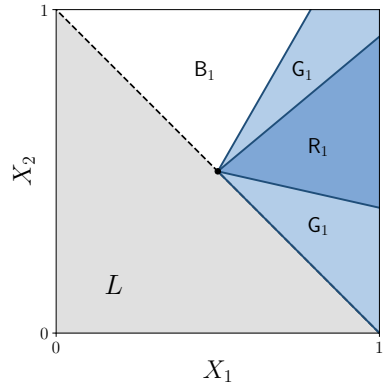
Differentiated Good News Signals

- Grade G reveals the borrower type but leaves the lender uncertain about competition
- Designate a core group of high-type borrowers for each lender: grade R
  - Make core groups non-overlapping
  - Ensure that each lender optimally charges borrowers in its core group a monopoly rate R



Differentiated Good News Signals

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  - Make core groups non-overlapping
  - Ensure that each lender optimally charges borrowers in its core group a monopoly rate R
- Strategic Segmentation: grades G and R both reveal a high-type borrower but induce different beliefs about the likelihood of competition
  - R competition is low
  - G competition is high

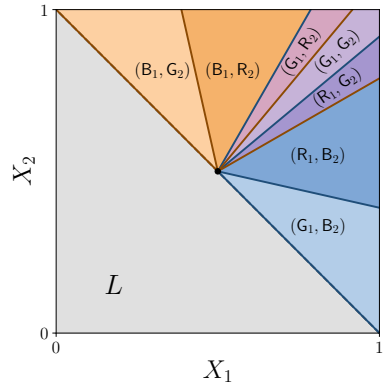


Optimal Differentiated Good News Signals

Proposition

Optimal differentiated good news signal satisfies:

- Monopoly pricing after R
- No lending after B
- No high type goes unfunded



Optimal Differentiated Good News Signals

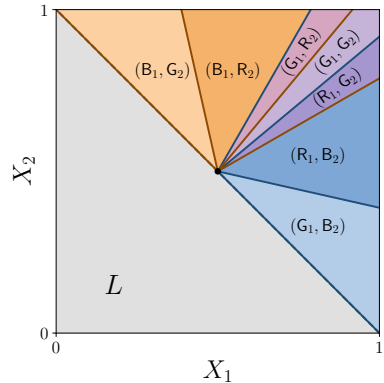
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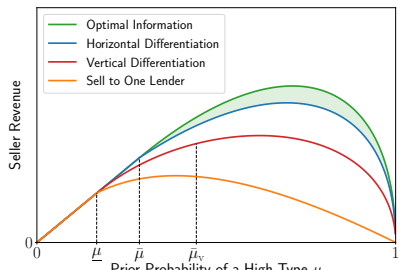
The information seller's revenue,

$$\mu(R-1)\big[1-\mathrm{P}(\mathsf{G},\mathsf{G}\mid H)\big].$$



Value of Strategic Segmentation

- For $\mu > \bar{\mu}$ deterrence comes at a cost
- Strategic segmentation further softens competition between lenders with positive (G and R) grades
- Weak competition entices lending even with B grades
- Higher precision $\alpha^o > \alpha^*$ restores deterrence



  - Optimal signal is more precise and has weaker effective competition

Revenue Bound and Optimal Information

- Optimizing over the space of all possible signals directly is not tractable
- Adopt the approach of Bergemann, Brooks and Morris (2017) to get an upper bound
  - Signals + bidding equilibrium $\sim$ Bayes Correlated Equilibrium
  - Only the winning rate matters $r_w = \min(r_1, r_2)$
    - Optimize over distributions of the winning rate
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Theorem

Optimal differentiated good news signal implements the upper bound on joint lender profits.

Extensions and Implications

Market with Many Lenders

Optimal Signal with Many Lenders

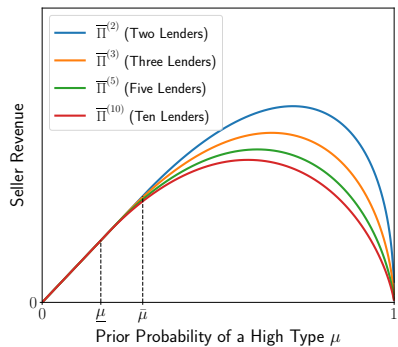
- As the number of lenders $N > 2$ increases
  - For a fixed precision, segmentation worsens
  - For a fixed segmentation, precision decreases and deterrence weakens

Proposition

Information seller's revenues decrease as the number of lenders goes up

$$\overline{\Pi}^{(N+1)} \leq \overline{\Pi}^{(N)},$$

and the differentiated good news signal remains optimal.

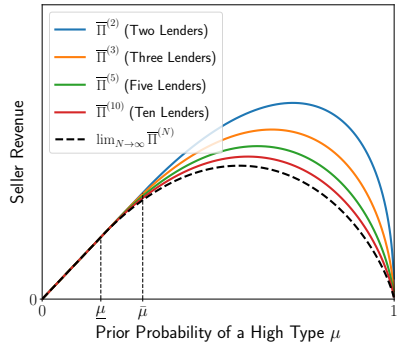


Limit with Many Lenders

- As the number of lenders $N > 2$ increases
  - Optimal precision decreases and deterrence weakens
- In the limit $\alpha^o \to 0$ and B grade becomes uninformative

$$\underline{R}_G^{(N)}(q_G^o) = \underline{R}(\mu_B(\alpha^o)) \to \underline{R}(\mu)$$

- Deterrence after B turns into deterrence of uninformed lenders



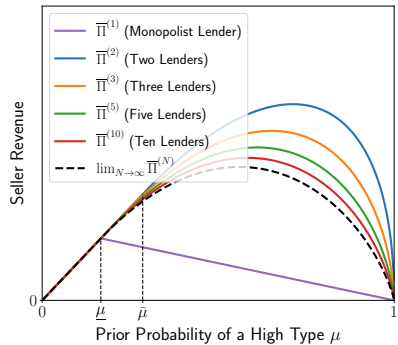
Proposition

Seller's optimal revenue converges to the revenue from selling information to only k of the N lenders while leaving the remaining ones uninformed $\overline{\Pi}_k^{(N)}$:

$$\lim_{n\to\infty} \overline{\Pi}^{(n)} = \overline{\Pi}_k^{(N)} \quad \text{for } N \geq 3 \text{ and } 2 \leq k \leq N-1.$$

Informing a Monopolist

- A monopolist lender who does not buy information could lend uninformed
- High outside option limits the ability of the seller to charge for information
- Seller can extract only screening value of information $(1-\mu)(1-pR)$

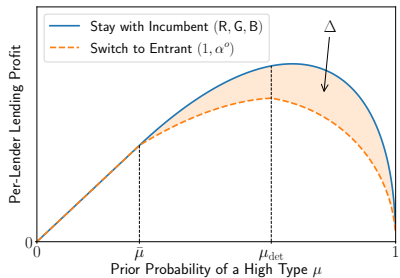


Competition Between Information Sellers

- An entrant information seller offers signals to lenders who decline the incumbent's offer
  - Exclusive purchases: each lender buys from one seller
  - Signals from different sellers are independent

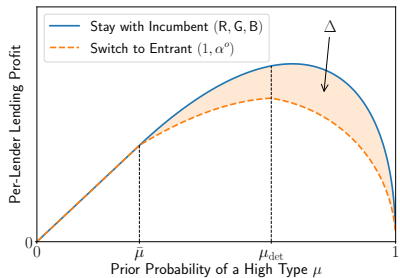
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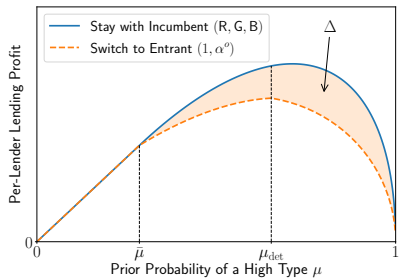
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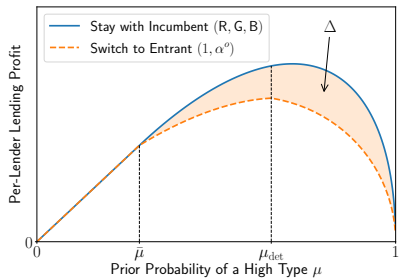
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- Switching from the incumbent to the entrant
  - Destroys strategic segmentation



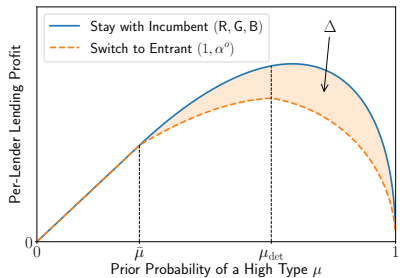
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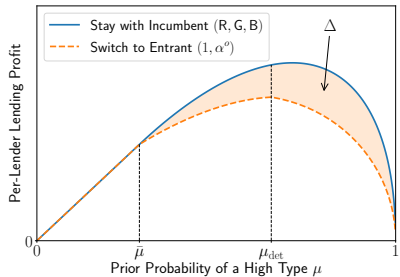
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- Incumbent's fee is the correlation premium $\Delta(\mu)$



Proposition

Entry leaves the incumbent's optimal signal and the lending equilibrium unchanged. Its revenue falls to $2\Delta(\mu)$, which is positive for $\mu > \bar{\mu}$.

Conclusion

1. Differentiated information softens lending competition
   - Deterrence: (noisy) low scores
   - Fundamental Segmentation: negative correlation of high scores
   - Strategic Segmentation: core group of high-quality borrowers for each lender

2. Vertically differentiated information increases welfare
   - Horizontally differentiated information does so only when μ is high
   - Strategic segmentation decreases borrower surplus without affecting welfare

3. More lenders increase borrower surplus
   - More information sellers might not

Thank you!